

Each formulation was applied in a 25 μ l dosage and a 50 μ l dosage for a total of 6 test groups. The receiver chamber was sealed to the atmosphere using parafilm wrap so that it was spill-proof and airtight. Three diffusion cells were used for each test group for a total of 18 cells.

[0350] The cells were filled with DI water for a receiver solution. The DI water had been degassed to remove air bubbles. The receiver solution was completely withdrawn and replaced with fresh DI water at each time point. Samples of the receiver solution were taken and analyzed by HPLC (high pressure liquid chromatography) to determine the leuprolide concentration. The cumulative amount of leuprolide across human cadaver skin was calculated from a 25 μ l and a 50 μ l solution containing NaOH, using the measured leuprolide concentrations in the receiver solutions for each time point (5 and 24 hours).

TABLE 44

Time	Cumulative Amount of Leuprolide (μ g/cm ²)					
	25 μ l solution			50 μ l solution		
	Leu-1	Leu-2	Leu-3	Leu-1	Leu-2	Leu-3
5 hours	0.38	0.52	0.58	0.32	0.62	0.3
24 hours	0.52	3.21	4.43	0.32	8.58	10.8

[0351] The cumulative amount of leuprolide across human cadaver skin for the 25 μ l dosage at 24 hours increased from 0.52 μ g/cm² to 4.43 μ g/cm² when the calculated sodium hydroxide concentration in the dried patch was increased from 0% to 6.7%. The cumulative amount of leuprolide across human cadaver skin for the 50 μ l dosage at 24 hours increased from 0.32 μ g/cm² to 10.8 μ g/cm² when the calculated sodium hydroxide concentration in the leuprolide solution was increased from 0% to 6.7%. The cumulative amount of leuprolide across human cadaver skin at 24 hours from the 50 μ l dosage group containing 3.6% NaOH (Leu-2) was 8.58 μ g/cm², which was about 27 times higher than that from the formulation without NaOH (0.32 μ g/cm², Leu-1).

[0352] The formulation of Leu-2 provided up to 6-fold (25 μ l solution) and to 27-fold (50 μ l solution) more leuprolide flux than in the absence of NaOH (Leu-1). The formulation of Leu-3 provided up to 9-fold (25 μ l solution) and up to 34-fold (50 μ l solution) more flux than in the absence of NaOH.

Example 14

[0353] The in-vitro permeation of oxytocin through human cadaver skin was performed using Franz-type diffusion cells with a diffusion area of 1 cm². The volume of receiver solution was 8 ml. Human cadaver skin was cut to a proper size and placed on a flat surface with the stratum corneum side facing up. The skin was clamped between the donor and receiver chambers of the diffusion cell. Eighteen diffusion cells were used in this study. A 2% NaOH aqueous solution (50 μ l) was introduced to the donor chambers of nine cells (cells #1 to 9) and a 4% NaOH aqueous solution (50 μ l) was introduced to the donor chambers of the other nine cells (cells #10 to 18). Once the NaOH solution is applied, the donor chamber was covered with parafilm.

[0354] After 5 hours, the NaOH solution was washed away from the skin for 3 cells (cells #1 to 3) that were treated with 2% NaOH solution and 3 cells (cells #10 to 12) that were treated with 4% NaOH solution. After 10 hours, the NaOH

solution was washed away from the skin for 3 cells (cells #4 to 6) that were treated with 2% NaOH solution and 3 cells (cells #13 to 15) that were treated with 4% NaOH solution. After 24 hours, the NaOH solution was washed away from the skin for 3 cells (cells #7 to 9) that were treated with 2% NaOH solution and 3 cells (cells #16 to 18) that were treated with 4% NaOH solution. To wash away the NaOH solution, the receiving fluid was removed and replaced with fresh DI water. This was done twice. DI water was added to the donor chamber to dilute the NaOH solution and then the donor solution was removed. This was repeated several times.

[0355] After the NaOH solution was washed away from the skin, the solution in the donor chamber was completely removed and replaced by 50 μ l of an oxytocin solution. The formulation of the oxytocin solution is listed in Table 45. Once the oxytocin solution was applied, the donor chamber was covered with parafilm.

TABLE 45

Formulation for the Oxytocin Solution	
Ingredient	g
Oxytocin	0.005
DI water	0.6
PG	0.6

The cells were filled with DI water as a receiver solution. The DI water had been degassed to remove air bubbles. The receiver solution was completely withdrawn and replaced with fresh DI water at each time point. The samples taken were analyzed by HPLC for the concentration of oxytocin in the receiver solution. The cumulative amount of oxytocin across human cadaver skin was calculated using the measured oxytocin concentrations in the receiver solutions for each time point, which were listed in Table 46. The skin was pretreated with 4% NaOH for the specified pretreatment time period.

TABLE 46

Time	Cumulative Amount of Oxytocin (μ g/cm ²)		
	5 hr Pretreatment	15 hr Pretreatment	24 hr Pretreatment
5 hours	118.95	202.28	193.82
15 hours	200.66	222.45	232.72
24 hours	225.52	231.58	236.80

Example 15

[0356] The in-vitro permeation of oxytocin through human cadaver skin was performed as described in Example 14, except that a 0.25% NaOH aqueous solution (50 μ l) was introduced to the donor chambers of nine cells (cells #1 to 9) and a 1.0% NaOH aqueous solution (50 μ l) was introduced to the donor chambers of the other nine cells (cells #10 to 18).

[0357] After 5 hours, the NaOH solution was washed away from the skin for 3 cells (cells #1 to 3) that were treated with 0.5% NaOH solution and 3 cells (cells #10 to 12) that were treated with 1.0% NaOH solution. After 11 hours, the NaOH solution was washed away from the skin for 3 cells (cells #4 to 6) that were treated with 0.25% NaOH solution and 3 cells (cells #13 to 15) that were treated with 1.0% NaOH solution. After 24 hours, the NaOH solution was washed away from the